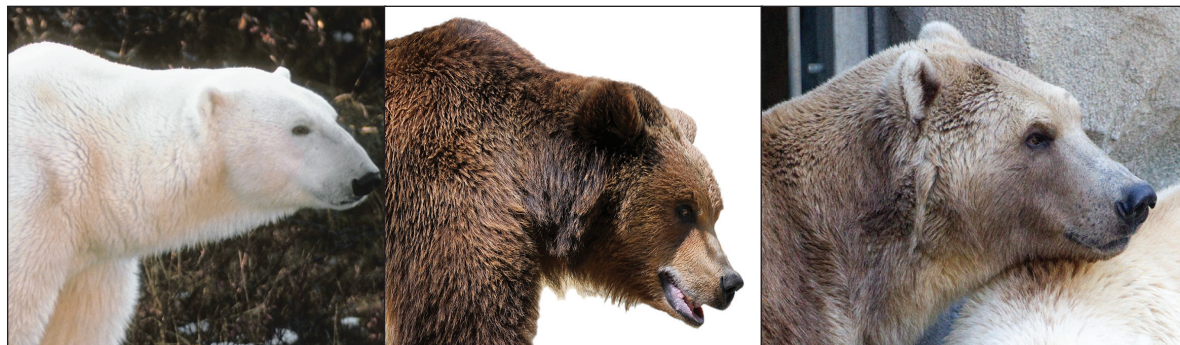


6. “Call it a ‘pizzly’ or a ‘grolar bear’, this new hybrid may be here to stay, say scientists. But is this the end to the polar bear as we know it or the start of a new bear species?”

This is the title of a recent scientific report stating that there is an increase in numbers of hybrid bears which are polar bear and brown (grizzly) bear hybrids and subsequent offspring which are  $\frac{3}{4}$  brown bear and  $\frac{1}{4}$  polar bear. The three types of bear are shown in **Image 6.1**.

**Image 6.1**



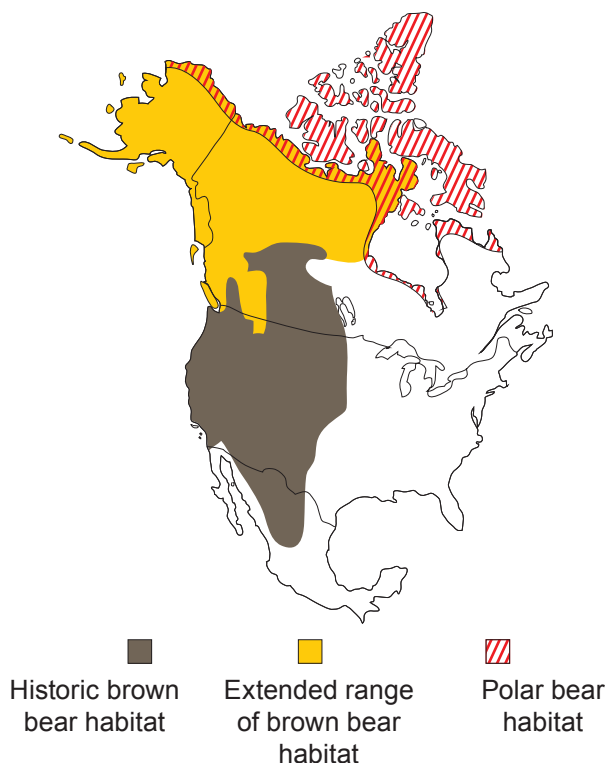
polar bear

brown bear

hybrid bear

The map shown in **Image 6.2** is the recently extended geographical range of the brown bear.

**Image 6.2**



At the start of the last ice age, a population of brown bears in the north became geographically isolated when increased ice stopped the free movement of the brown bears. Following their isolation, these brown bears evolved to become polar bears.

Brown bears eat tubers, berries and fish. Polar bears rapidly evolved a metabolism that let them live off a blubber-rich diet, which would be unhealthy for most mammals. Polar bears have genes related to fat transport, fatty acid metabolism, and cardiovascular function – all of which helps them survive on a diet of mostly fatty marine mammals like seals. Polar bears adjusted to this ultra-high-fat diet in less than 20 500 generations.



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Since 2008, brown bears have been spotted further north than their historic range, a possible consequence of climate change. As a result, the brown bear habitat now overlaps the polar bear habitat. The two kinds of bears, which are capable of mating and producing viable offspring, are considered to be separate species.

Describe how the process of geographical isolation has resulted in the evolution of polar bears from the brown bear.

Explain how the polar bear has adapted to live in its habitat.

Suggest how the existence of hybrid bears affects scientists' confidence that they are two different species. [9 QER]

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This image shows a single page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

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**END OF PAPER**

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